INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

GRC205: Incident Management and Business Continuity
Week 21 Assignment (30%)

BUSINESS CONTINUITY AND DISASTER RECOVERY REPORT
Northbridge Skills Institute: Flood and Power Failure

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Introduction

This report explains how Northbridge Skills Institute can continue working and recover its systems after flooding and a power failure. The incident starts at 08:00 on Monday, Day 0, and the main site is expected to be closed for about 72 hours. Microsoft 365 email and identity services still work, but local systems are unavailable. The figures in this report are estimates for the exercise, not actual losses.

Assumptions: The examination lasts two hours; two fair versions of the exam can be prepared; each restored system needs a 30-minute business check before it is declared recovered; the alternate facility has power and internet; and no cyberattack is confirmed at the start. The roles named in the report are proposed roles for the institute.

Task 1: Business Impact Analysis

1.1 Main business functions

A Business Impact Analysis (BIA) identifies the services that matter most, who depends on them, and what happens when they stop. Priority 1 means the most urgent service.

Function and owner	People affected / deadline	Main impact	Minimum service during outage	Priority
Teaching and exams Academic Director / Exams Officer	1,800 learners; 120 exam candidates; exam at 11:00	Exam delay, lost learning, stress and complaints	Move classes to Teams; run a controlled exam at an approved alternate site or online; check IDs and keep attendance records	1
Fees and finance Manager	Learners, finance staff and sponsors; fee confirmation by 16:00	Missing or incorrect receipts and payment disputes	Check the payment provider portal; use numbered receipts; require two-person approval; do not issue manual refunds	2
Admissions Manager	Applicants and current learners; admissions close at 18:00	Lost applications and damage to reputation	Use a restricted Microsoft 365 form and numbered log; enter records into REG01 after recovery with a second-person check	3
Payroll HR Manager and Finance Manager	90 staff; bank deadline at 12:00 on Day 1	Late or incorrect pay	Use verified HR records; if needed, use the last	4

			approved payroll file plus approved changes; require two approvals and check for duplicate payments	
Records and certificates Records Manager	Graduates, employers and other institutions; 48-hour service target	Incorrect certificates or exposure of private information	Log every request; do not issue certificates until file integrity and access rights are checked	5

1.2 Estimated financial impact

If no recovery or temporary measures are used, the estimated cost is:

Function	Hourly cost (NGN)	8-hour cost (NGN)	24-hour cost (NGN)
Teaching and exams	60,000	600,000 (including 120,000 exam cost)	1,560,000 (includes 120,000 exam cost)
Fees and finance	40,000	320,000	960,000
Admissions	20,000	160,000	480,000
Payroll	10,000	80,000	240,000
Records and certificates	5,000	40,000	120,000
Total	135,000	1,200,000	3,360,000

These figures are estimates. Real costs may change depending on the time of day, extra staff hours, recovery costs and how much work can be completed later. Money is not the only concern. Learner safety, fair exams, the institute's reputation and correct records are also important.

1.3 Priorities and system dependencies

The exam is the first concern because it starts at 11:00. Fees must be confirmed by 16:00, and admissions close at 18:00. Payroll is due the next day, while records have a 48-hour target.

Systems must be restored in the correct order. First restore the recovery network, then AUTH01 (identity and login), then check security and backups. Next restore DB01 and FS01. After DB01 is ready, restore the main applications, including LMS01, FIN01, REG01 and HR01.

Main risks include DB01 being needed by several functions, lost account changes after midnight, one administrator account controlling both live systems and backups, only two technicians being available, and dependence on outside suppliers. These risks must be checked during recovery.

Task 2: Recovery Objectives and Backup Design

2.1 Recovery targets

The Recovery Time Objective (RTO) shows how quickly a service should be restored. The Recovery Point Objective (RPO) is how much recent data the institute can afford to lose. Maximum Tolerable Downtime (MTD) is the longest time a service can remain unavailable before the harm becomes unacceptable. The RTO should be shorter than the MTD.

Function	MTD	RTO	RPO	Reason
Teaching and exams (F1)	24 hours	5 hours	1 hour	Teaching should not stop for more than a day; exam requires a temporary plan.
Fees and finance (F2)	8 hours	5 hours	1 hour	Allows time to confirm payments before the 16:00 deadline.
Admissions (F3)	10 hours	6 hours	1 hour	Allows applications to be collected and entered later.
Payroll (F4)	28 hours	24 hours	4 hours	Allows time to check and approve payroll before the bank deadline.
Records (F5)	48 hours	24 hours	4 hours	Correct records and certificates are more important than speed.

A service is only counted as recovered when the system works, the data checks pass, and the service owner confirms that staff can use it.

2.2 Current backup gaps

The database backup chain can recover data up to 07:00, so it meets the one-hour RPO only if the chain is complete. If the chain fails, the latest full backup is from 23:00 the previous night, leaving a nine-hour gap. If compromise is suspected, the disconnected backup from three days earlier may be needed, causing a much larger data loss.

The AUTH01 snapshot is from midnight, so password changes and disabled accounts made after midnight must be checked and applied again. The FS01 full backup is from 23:00 the previous night, which leaves a nine-hour gap and does not meet its four-hour RPO. Microsoft 365 forms and emails, payment provider reports, paper copies and approved HR emails can help rebuild some missing information.

2.3 Improved backup plan

System	Backup plan	Retention and protection
DB01 database	Weekly full backup; daily differential backup; transaction log backup every 15 minutes from 06:00 to 22:00	Keep logs 14 days, daily copies 35 days, weekly copies 13 weeks and monthly copies 12 months.
AUTH01 identity	Daily system-state backup and a snapshot every four hours	Keep for 35 days; also keep an offline directory export.
FS01 files	Weekly full backup and incremental backup every four hours from 07:00 to 19:00	Keep daily copies 35 days and monthly copies 12 months.
Applications	Weekly system image and a configuration backup after each change	Keep copies for 13 weeks.

Keep at least three copies of important data: the live copy, a warm-site copy and an offsite copy. Use at least two types of storage and keep one copy offline or locked so it cannot easily be changed or deleted. Encrypt all copies, use a separate backup administrator account with multi-factor authentication, and keep encryption keys separate from the main identity system.

A backup is useful only if it can be restored and checked. Test file hashes, database consistency, record counts, last transaction times, sample logins and sample transactions. An offsite copy or live replica alone is not enough because it may contain damaged or incorrect data.

Task 3: Recovery Strategy and Budget

3.1 Compare the options

Option	Recovery time	Main advantage	Main weakness	Annual total
Cold site	24–48 hours	Lowest cost and less dependence on live replication	Too slow for the first-day deadlines	NGN 1,200,000
Warm site	3–5 hours	Balances speed, cost and recovery from checked backups	May not be fast enough for the 11:00 exam	NGN 2,100,000
Hot site	About 1 hour	Fastest recovery and best chance of meeting tight targets	Highest cost and may copy corruption quickly	NGN 3,000,000

Each total includes the base package, NGN 400,000 for protected backups and NGN 200,000 for training and tests. The annual budget limit is NGN 3,000,000.

3.2 Recommendation

I recommend the warm-site option at NGN 2,100,000 per year. It leaves NGN 900,000 under the budget limit and can meet most recovery targets if the recovery work goes well and the improved backup plan is used. It also restores from checked backups rather than relying only on a live replica.

The warm site cannot save the 11:00 exam because it takes several hours to restore the systems. The institute should approve an exam workaround in advance. The cold site is too slow for the first-day deadlines. The hot site costs the full budget and leaves no money for unexpected costs.

Before final approval, the institute should review previous floods and power failures, other risks such as ransomware and fire, supplier contracts and test reports, supplier security evidence, and the results of its own recovery tests. The principal should formally accept any remaining risks.

Task 4: Business Continuity and Disaster Recovery Plan

4.1 Activation and leadership

Activate the plan if the main site is expected to be unavailable for more than two hours, a critical system is down for more than one hour, or an important deadline is at risk. Signs of a cyberattack must also trigger the incident response team.

The principal makes the final decision. If the Principal cannot be reached within 15 minutes, the Deputy Principal for Operations acts as deputy. The BC/DR Coordinator runs the plan and advises the principal. Facilities must report the incident immediately. The coordinator briefs the principal by 08:30, and the activation decision should be made by 08:45.

Use Microsoft 365 email and Teams over mobile internet as the main communication methods. SMS and WhatsApp are backups. Keep a printed call tree for emergencies. Situation reports should be sent at 09:30, 11:00, 13:00, 16:00 and 18:00 on Day 0, then at 09:00 and 17:00 each day.

4.2 Roles and responsibilities

Role	Main responsibility
Principal	Approves activation, accepts major risks and gives final sign-off.
BC/DR Coordinator	Coordinates the teams, sends updates and maintains the plan.
DR Lead and technicians	Check backups and restore systems.
Function owners	Check that restored services and data are correct.
Exams Officer	Organises exam changes and exam controls.
Facilities Manager	Checks building safety and arranges the alternate site.
Incident Response / Security Lead	Investigates signs of compromise and decides which backups are safe.
Communications Officer	Sends clear updates to learners, staff and other stakeholders.

4.3 Temporary procedures during the 72-hour closure

Teaching should continue through Microsoft Teams. Since the exam system will not be ready by 11:00, the Principal and Academic Director should approve new exam times and inform all candidates. The alternate facility has 25 workstations. Use different exam versions, check learner IDs, record attendance, control phones and provide accessible seats and extra time where needed. Online exams should have suitable invigilation.

Finance staff should check payments through the payment provider portal and use a numbered receipt register. Two people must approve each entry. Do not issue manual refunds or store card details. Admissions staff should use a restricted Microsoft 365 form and numbered log, then enter the records into REG01 after recovery with a second-person check.

For payroll, use HR01 once it has been checked. If it is unavailable, use the last approved payroll file plus authorised changes. HR and Finance must both approve it and check with the bank to prevent duplicate payments. For certificates, record each request but do not issue a certificate until FS01 data and access rights are checked.

4.4 Recovery runbook

Step	Action	Check before moving on
1	Confirm activation and choose the required backups.	Record the backup list and hashes.
2	Restore the recovery network.	Test routing and firewall rules.
3	Restore AUTH01.	Test logins, reset administrator passwords and disable staff accounts that should no longer work.
4	Check backup integrity and scan for malware.	Confirm hashes, review administrator logs and involve Incident Response if anything looks suspicious.
5	Restore DB01 and check the database.	Confirm consistency, record counts and the last transaction time.
6	Restore FS01.	Check file hashes and access permissions.
7	Restore LMS01, FIN01, REG01 and HR01.	Test logins and sample transactions.
8	Ask each service owner to approve the restored service.	Keep manual procedures running if checks fail.
9	Tell users the service is available and update the incident report.	If the service fails, take it offline and return to the manual process.

4.5 Communication and safe return

Tell learners by email and SMS that the main campus is closed, that they should not travel there, and that exam and deadline changes will be shared. Tell staff where to work and what duties they have.

Contact the payment provider, bank, alternate facility, recovery providers and awarding body as needed. If a data breach is suspected, contact the data protection lead immediately.

Do not return to the main site until Facilities confirms in writing that the building is safe, electrical systems are certified, power has been stable for 24 hours and equipment has been checked for water damage. Back up new data created at the warm site, transfer it carefully and compare records and hashes. Each service owner must test their service. Keep the warm site available for seven days in case the institute needs to return to it.

Task 5: Recovery Schedule and Exercise

5.1 Planned recovery schedule

Action	Start	Finish
Recovery network	08:00	08:30
AUTH01	08:30	09:00
Security and backup checks	09:00	09:30
DB01	09:30	11:00
FS01 (second technician)	09:30	10:30
LMS01	11:00	11:45
FIN01	11:00	11:30
HR01	11:30	12:15
REG01	11:45	12:15

The planned schedule shows that the exam system, LMS01, is technically restored at 11:45 and accepted at 12:15 after the 30-minute check. Therefore, the 11:00 exam cannot start on time and must use the temporary plan. The other services can be recovered within their planned RTOs if every backup and recovery step succeeds.

5.2 Tabletop exercise and lessons learned

A tabletop exercise is a discussion-based test of how the team would respond. In the exercise, the closure was extended to five days, the alternate site had only 15 working computers, a database backup failed its check, and some payments were missing from the restored finance records.

The team decided to extend the alternate site and warm-site arrangements, continue teaching online, and arrange staff rest periods. With only 15 computers, the exam was split into three sittings and three paper versions were prepared. When the database backup failed its check and an unexpected administrator login was found, the team stopped the restore, involved Incident Response, disabled the shared administrator account and chose the last verified backup. The finance team used the payment provider's report to check missing payments and told learners not to pay again.

Corrective action	Priority	Owner	Target date
Separate backup administration from production and require MFA.	High	IT Manager	30 Nov 2026

Add 15-minute database log backups and a 30-day immutable copy.	High	DR Lead	31 Dec 2026
Document on how the offsite provider protects backup access.	High	Procurement Officer	31 Jan 2027
Approve an alternative exam plan and arrange laptops for 40 learners.	High	Exams Officer	15 Jan 2027
Reconcile payment records with provider reports every day.	Medium	Finance Manager	31 Jan 2027
Add four-hourly AUTH01 snapshots and FS01 backups.	Medium	DR Lead	31 Dec 2026
Test recovery from the disconnected backup.	Medium	DR Lead	28 Feb 2027

Task 6: Governance and Performance Measures

6.1 Ownership and testing

The Principal owns the plan and approves it every year. The governing board's risk committee reviews the main risks. The BC/DR Coordinator maintains the plan, the IT Manager maintains the technical runbook, and each service owner maintains their temporary procedures.

When	Test or activity	Purpose
Monthly	Restore a sample system and check its data.	Check that backups work and meet RPO targets.
Quarterly	Run a tabletop exercise with new problems.	Check decisions, roles and communication.
Every March, before the rainy season	Test warm-site recovery of AUTH01, DB01 and LMS01.	Check recovery times and flood readiness.
Before each exam period	Review the alternative exam plan.	Make sure the exam can continue fairly.
Annually	Review and approve the plan and train staff.	Keep the whole plan current.

Review the plan after an incident, a failed test, a major system or supplier change, a change in key staff, a new legal requirement or a serious flood warning. Keep a controlled digital copy, an encrypted offline copy and printed copies with key leaders. Remove old versions when a new one is approved.

6.2 Frameworks and flood risks

The report follows the main steps of NIST SP 800-34: setting policy, carrying out a Business Impact Analysis, choosing controls, selecting recovery strategies, writing the plan, testing it and maintaining it.

It also supports ISO 22301 ideas about leadership, operations, performance checks and improvement. This mapping is for learning purposes and does not mean the institute is certified.

Flooding is a major risk for Northbridge. The warm site and alternate facility should be in a different flood area from the main campus. Server and power equipment should be raised above likely flood levels. The institute should monitor flood warnings and run its full recovery test before the rainy season.

6.3 Key performance indicators

Measure	Target	How often
Functions recovered within RTO	100% for F1–F3; 90% overall	Quarterly
Critical systems with a successful restore test in the last 90 days	100%	Quarterly
Corrective actions completed by their due dates	At least 90%	Monthly
Role holders trained and tested within the last 12 months	100%	Quarterly
Days when the verified DB01 backup is no older than one hour	At least 99%	Monthly

6.4 Maturity assessment

Northbridge is at Level 2, Developing. It has backups, an offsite copy, a warm-site option and some recovery timing results. However, backup access is not fully separated, supplier independence has not been proven, some backups do not meet the required RPO, the exam workaround is not yet approved, and regular restore testing is missing.

The target is Level 3, Defined, within 12 months. To reach it, the institute should approve the plan, complete all seven corrective actions, put the improved backup system in place, complete a full failover test and report its performance measures every quarter for at least two quarters. Level 4 should be considered after the institute has a longer record of good results.

Conclusion

Northbridge should choose the warm-site recovery option and improve its backup protection. The institute must use temporary procedures to protect exams, fees, admissions, payroll and records while systems are unavailable. Recovery must include security checks and approval from service owners, not just switching servers back on. Regular tests, clear responsibilities and safer backup access will help Northbridge recover more safely from future floods, power failures or cyber incidents.

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